

“Cross-National Analysis of the Determinants of Later-Life Cognitive Performance in the United States, the Caribbean, and Mexico”

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ABSTRACT

INTRODUCTION

Cross-national differences in later life cognitive performance may reflect unequal exposure to life-course, health, and sensory risk factors for cognitive aging.

METHODS

We analyzed harmonized cross-national data for Hispanic adults over 65 in the United States and the Dominican Republic, Cuba, Puerto Rico, and Mexico. Weighted multivariable regressions and Oaxaca-Blinder decompositions examined TICS-27, delayed recall, and animal naming.

RESULTS

U.S. Hispanics had higher mean scores than the Caribbean/Mexico group. Observed characteristics explained 36% of the TICS-27 gap, 32% of the delayed-recall gap, and 41% of the animal-naming gap. Education was the dominant explained component, especially for the Dominican Republic and Mexico and the gaps in Cuba and Puerto Rico were predominantly unexplained. Hearing was more consistently associated with cognition than vision.

DISCUSSION

Educational opportunity and modifiable health conditions are important correlates of cross-setting cognitive disparities, but measurement, schooling quality, and broader contextual factors require further study.

1. Introduction

Cognitive impairment and dementia are major contributors to disability, dependence, and healthcare need in later life. However, their burdens are distributed unequally across populations and countries. Contemporary dementia prevention frameworks highlight that cognitive health reflects cumulative exposure to social and health conditions across the life course, including education systems and opportunities, cardiometabolic disease, depression, and sensory impairment¹. Thus, understanding population differences in later life cognitive performance requires more than comparing mean test scores but it requires identifying whether observed gaps reflect differences in the distribution of these characteristics or whether they persist among populations with otherwise similar measured characteristics.

Prior cross-national research has shown substantial variations in cognitive performance among older adults³. Education has been documented as one of the most consistent and dominant determinants of later life cognition, but it may represent different educational exposures across countries⁶. The same number of completed educational years may also have systematically different returns due to instructional quality, adherence and dropouts, curriculum, and subsequent economic opportunities. For instance, study in Puerto Rico indicates that measures of educational quality provide information about late-life cognitive functioning beyond years of schooling alone⁷. Furthermore, chronic disease, depression, and sensory impairment may also significantly contribute to cognitive differences, but their prevalence, diagnosis, treatment, and measurement can vary across healthcare systems¹.

Evidence remains particularly limited for older adults in the Hispanic Caribbean. Prior studies including the U.S. Health and Retirement Study (HRS) and the Mexican Health and Aging Study (MHAS) have documented population level research on aging and cognition in the United States and Mexico^{8,9}. In contrast, comparable population based cognitive aging data from the Dominican Republic, Cuba, and Puerto Rico have historically been sparse^{4,5}. The Caribbean American Dementia and Aging Study (CADAS) now provides harmonized cognitive and health information from these three Caribbean countries⁴. This opens a valuable opportunity to investigate the variations across Hispanic and Latino populations that share some historical, cultural, and linguistic features but differ substantially in educational systems, life-course health risk factors, and access to health care.

Existing cross-national studies have generally focused on documenting differences in cognitive levels or age patterns yet less is known about how much of the observed variation between U.S. Hispanics and older adults in the Hispanic Caribbean and Mexico is associated with observable differences in education, demographic composition, chronic disease, depression, vision and hearings. We address these gaps by using harmonized cross-sectional data for adults over 65 or older from the U.S. Health and Retirement Study, the Caribbean American Dementia and Aging Study, and the Mexican Health and Aging Study. Our primary cognitive outcome measures are TICS-27, delayed word recall, and animal naming. We first describe weighted differences in cognitive performance across settings and estimate adjusted associations with demographic, educational, health, and sensory characteristics. We then use Oaxaca-Blinder decomposition to quantify the portions of mean cognitive gaps associated with observed population characteristics and the remaining

unexplained components by those characteristics¹²⁻¹⁴. In addition, we also examine the role of nativity (Foreign-born vs US-born Hispanics) within the HRS sample. By distinguishing observed compositional differences from residual cross-setting disparities, this study clarifies both the importance and the limitations of education and modifiable health characteristics in explaining cognitive variation across diverse Hispanic and Latino populations.

2. Method

2.1 Data sources and study population

For the analysis, we used harmonized data from the HRS in the United States, CADAS in the Dominican Republic, Cuba, and Puerto Rico, and MHAS in Mexico. HRS is a nationally representative longitudinal survey data of U.S. adults older than 50 years⁸. The present HRS analysis was restricted to respondents who identified as Hispanic. CADAS was designed to characterize cognitive aging and dementia in Caribbean populations using harmonized methods. Its baseline samples are nationally representative in the Dominican Republic and Puerto Rico and population-based in selected urban and rural areas of Cuba⁴. MHAS is a nationally representative longitudinal survey data of adults aged 50 years or older in Mexico and was designed to be broadly comparable with HRS⁹.

We used the 2022 HRS wave, baseline CADAS data, and the 2021 MHAS wave. For HRS respondents without an observed 2022 cognitive measure, an available 2020 measure was used according to the prespecified harmonization procedure. We restricted each study sample to direct respondents who are 65 old or older and complete information for the relevant cognitive outcome

and the covariates. Because cognitive outcomes availability differed slightly, analytic sample sizes were outcome specific.

2.2 Cognitive Outcomes

The primary outcome was TICS-27, a global cognitive score constructed from immediate and delayed word recall, serial-7 subtraction, and backward counting commonly used in HRS studies^{10,11}. TICS-27 scores range from 0 to 27, with higher scores indicating better cognitive performances. The two domain specific outcomes were delayed recall, which is the number of words remembered after a delay, and animal naming which is the number of animals named during a timed semantic fluency task. Items were aligned across studies so that higher values consistently represented better performance. We evaluate continuous test performance rather than assigning clinical dementia categories for quantitative analysis.

2.3 Covariates

Covariates were selected based on established cognitive-aging risk factors and cross-study availability. Demographic variables included age in years and gender. Education was measured as completed years of schooling and health variables included indicators for diabetes, hypertension, and binary CES-D scale for depression measure. Sensory functions were measured with self-rated vision and hearing. Each was represented by indicators for excellent, very good, good, and fair function, with poor function as the reference category.

2.4 Statistical Analysis

We first calculated weighted means, standard deviations, and proportions by study setting using survey weights. We plotted weighted quadratic predictions of each cognitive outcome over

age 65 to 100, with 95% confidence intervals, to summarize age patterns without imposing a common linear slope across settings.

We next estimated weighted multivariable linear regressions for TICS-27, delayed recall, and animal naming to investigate the association between cognitive outcomes and covariates. The main presentation contrasted the HRS Hispanic sample with the combined CADAS/MHAS sample. Supplementary models were estimated separately for each setting in the Appendix.

Next, we used Oaxaca-Blinder decompositions to partition weighted mean differences into an explained component and an unexplained component¹²⁻¹⁴. The explained component represents the part associated with group differences in the distributions of observed covariates, evaluated using an equally weighted combination of the two groups' coefficients as the reference structure (weight = 0.5). The unexplained component shows the differences in coefficients and the intercept, together with contributions from unmeasured factors. It should not be interpreted as a causal effect of country, discrimination, or a direct estimate of unobserved disadvantage.

Primary decompositions compared HRS Hispanics with the pooled Dominican Republic, Cuba, Puerto Rico, and Mexico group and the pooled three CADAS Caribbean samples with Mexico. Country-specific decompositions compared HRS with each non-HRS setting. Nativity analyses compared foreign-born HRS Hispanics with U.S. born HRS Hispanics and foreign-born HRS Hispanics with the pooled CADAS/Mexico sample. Robust standard errors were clustered at the respondent level. Analyses were conducted in Stata using the `oaxaca` command. The analysis is done in Stata BE version 18.5.

3. Results

The TICS-27 decomposition sample included 1,063 HRS Hispanics, 1,450 participants from the Dominican Republic, 1,421 from Cuba, 705 from Puerto Rico, and 2,596 from Mexico (Table 1). Weighted mean age ranged from 72.6 years in Mexico to 75.9 years in Cuba. Women represented 56% to 65% of each sample. Educational attainment varied substantially. Mean years of schooling were 10.0 in HRS, 5.1 in the Dominican Republic, 11.0 in Cuba, 11.9 in Puerto Rico, and 4.9 in Mexico. Reported health conditions also differed as diabetes prevalence was highest in HRS (45%), hypertension was common in all settings (53% to 73%), and depression was most prevalent in Mexico (45%). Self-rated vision and hearing distributions varied significantly, emphasizing the need for cautious cross-setting interpretation of sensory categories.

Weighted mean cognitive performance was highest in the HRS Hispanic sample for all three outcomes (Table 2). Mean TICS-27 was 12.39 in HRS, compared with 8.64 in the Dominican Republic, 10.85 in Cuba, 8.58 in Puerto Rico, and 9.08 in Mexico. Mean delayed recall was 3.76 in HRS, 2.48 in the Dominican Republic, 3.32 in Cuba, 2.37 in Puerto Rico, and 2.74 in Mexico. Mean animal naming was 15.50 in HRS, 11.52 in the Dominican Republic, 14.00 in Cuba, 13.15 in Puerto Rico, and 13.75 in Mexico. Figure 1 showed declining predicted performance with age in every setting and outcome, with some curvature and wider confidence intervals at the oldest ages. Cuba was generally closest to HRS, whereas the Dominican Republic and Puerto Rico had lower TICS-27 and delayed-recall profiles.

In adjusted regressions, age was negatively associated with all outcomes in both HRS and the pooled CADAS/Mexico sample (Table 3). Education was the most consistent positive correlate. Each additional year of schooling was associated with 0.329 higher TICS-27 points in HRS and 0.301 points in CADAS/Mexico and corresponding associations were 0.105 and 0.085 for delayed recall and 0.362 and 0.238 for animal naming. Depression was associated with lower TICS-27 and

delayed recall in both groups and with lower animal naming in the pooled CADAS/Mexico group. Diabetes was negatively associated with TICS-27 and delayed recall in CADAS/Mexico, while hypertension associations were less consistent. Female sex was positively associated with delayed recall but negatively associated with animal naming in CADAS/Mexico, indicating domain-specific rather than uniformly favorable or unfavorable differences.

Hearing showed more consistent adjusted associations with cognition than vision. Relative to poor hearing, very good or good hearing was associated with higher TICS-27 and delayed-recall scores in both HRS and CADAS/Mexico. Vision coefficients were heterogeneous across outcomes and settings and did not follow a uniform gradient. These findings support consideration of sensory function but do not establish that differences in self-reported sensory status causally produced the observed cognitive gaps.

In the primary decomposition, the mean HRS advantage over the pooled Caribbean/Mexico group was 3.061 points for TICS-27, 1.002 point for delayed recall, and 2.217 points for animal naming (Table 4). Observed characteristics explained 1.100 TICS-27 points (35.9% of the gap), 0.319 delayed-recall points (31.8%), and 0.907 animal-naming points (40.9%). Education was the largest explained component for every outcome which is, 0.904 points for TICS-27, 0.271 for delayed recall, and 0.864 for animal naming. Differences in age, depression, diabetes, and hearing made smaller contributions. The remaining unexplained components were 1.961, 0.683, and 1.310 points, respectively (all $p < 0.01$).

Comparisons of the pooled CADAS Caribbean samples with Mexico showed smaller and less uniform mean differences. The Caribbean group scored 0.431 points higher on TICS-27, was not significantly different in delayed recall, and scored 0.989 points lower on animal naming. Although education strongly favored the Caribbean group in the explained component, the total

unexplained component offset or reversed that advantage for several outcomes. This results illustrate why the explained and unexplained components should be interpreted jointly rather than treating the explained component as a prediction of the observed gap.

Country-specific decompositions show substantial heterogeneity. For HRS and the Dominican Republic comparison, observed characteristics explained 1.811 of the 3.744-point TICS-27 gap (48%), 0.637 of the 1.301-point delayed-recall gap (49%), and 1.450 of the 3.979-point animal-naming gap (36%) and education accounted for most of each explained component. For HRS and Mexico comparison, observed characteristics explained 1.993 of the 3.310-point TICS-27 gap (60%), 0.561 of the 1.018-point delayed-recall gap (55%), and 1.548 of the 1.650-point animal-naming gap (94%), again largely through education. In contrast, the explained components for Cuba were small and not statistically significant, and the explained components for Puerto Rico were negative. Because mean schooling was higher in Cuba and Puerto Rico than in HRS, education differences predicted narrower or reversed gaps, while the observed cognitive gaps remained positive. Consequently, the HRS-Cuba and HRS-Puerto Rico differences were predominantly unexplained by the included covariates.

Nativity analyses added a life-course perspective (Table 5). Within HRS, foreign-born and U.S.-born Hispanics did not differ significantly in TICS-27 or delayed recall (Although foreign-born Hispanics score lower in both), but foreign-born participants scored 1.491 points lower on animal naming. Differences in education predicted lower scores among foreign-born participants for all three outcomes, although unexplained components offset much of the predicted disadvantage in TICS-27 and delayed recall. Compared with the pooled CADAS/Mexico group, foreign-born HRS Hispanics scored 2.870 points higher on TICS-27, 0.958 points higher on delayed recall, and 1.661 points higher on animal naming. Observed characteristics explained

0.793, 0.224, and 0.573 points of these respective gaps, leaving statistically significant unexplained components of 2.077, 0.734, and 1.088 points.

Discussion

In this harmonized comparison of older adults across the United States, the Hispanic Caribbean, and Mexico, HRS Hispanics had scored higher in TICS-27, delayed-recall, and animal-naming than the pooled CADAS/Mexico group. Observable demographics, education, health, and sensory characteristics explained approximately one-third to two-fifths of the pooled gaps. Education was by far the largest explained component and the most consistent adjusted correlate of cognition. Furthermore, the extent of explanation differed sharply by setting as education accounted for a large share of the HRS-Dominican Republic and HRS-Mexico gaps, whereas differences involving Cuba and Puerto Rico were predominantly unexplained despite even though they have higher average years of schooling.

The dominant effect of education is consistent with evidence that schooling can support cognitive reserve, shape occupational and economic opportunities, improve health literacy, and influence cumulative exposure to cognitive stimulation^{1,3,6}. However, the findings also show the limitations of using years of schooling as a complete measure of educational advantage. In Cuba and Puerto Rico, participants reported more schooling on average than HRS Hispanics, so the compositional education term predicted that they should perform similarly or better. Their observed mean scores, particularly for TICS-27 and delayed recall, remained lower. Years completed may therefore be capturing different levels of instructional quality, literacy, school adherence or dropouts, curriculum differences, language exposure, or socioeconomic return across cohorts and settings. Prior work in Puerto Rico has shown that indicators of educational quality

can add information beyond years of schooling such that incorporating literacy, school location, childhood socioeconomic conditions, and occupation would help distinguish these pathways⁷.

Depression and diabetes were also associated with lower cognitive performance in several models, consistent with their roles in contemporary dementia-prevention frameworks¹. Their decomposition contributions were smaller than those of education. Hypertension was not consistently associated with the outcomes, which may reflect treatment, survival, diagnosis, or the limitations of a cross-sectional self-reported indicator. Cross-setting differences in diagnostic access and disease awareness also complicate direct comparison of reported chronic conditions.

Hearing was more consistently associated with TICS-27 and delayed recall than vision. Hearing impairment may affect communication during testing and may also mark longer-term reductions in social and cognitive engagement^{1,15}. Nevertheless, the sensory results should not be read as causal. Self-rated sensory categories may not be equivalent across cultures, and test administration can magnify the association between hearing and performance even when hearing is not the underlying cause of cognitive differences. Objective audiometric and visual measures, information on assistive-device use, and standardized administration procedures would strengthen future work. The heterogeneous vision coefficients similarly counsel against interpreting a single category contrast as a graded protective effect.

The nativity results indicate that Hispanic ethnicity does not represent a single educational or cognitive-aging context. Within HRS, foreign-born participants had fewer years of schooling than U.S.-born participants, and these educational differences predicted lower performance across all three outcomes. Nevertheless, foreign-born HRS Hispanics performed better on average than the pooled CADAS/Mexico group. Most of these cross-setting differences, particularly for TICS-27 and delayed recall, remained unexplained by the included demographic, health, sensory, and

educational measures. Schooling quality, language and test familiarity, migration selection, age at migration, occupational histories, and health-care access are possible explanations, but these mechanisms were not measured directly.

This study has several strengths. It uses newly available CADAS data to broaden comparative cognitive-aging research in the Hispanic Caribbean, examines both a global score and two domain-specific outcomes, applies sampling weights, and separates observed compositional differences from residual differences using a Oaxaca-Blinder decomposition framework. The country-specific analyses also represent the important heterogeneities across countries.

However, there are several limitations. First, the analysis is cross-sectional and cannot identify causal effects or distinguish level differences from differences in cognitive decline. In addition, survey waves were not perfectly contemporaneous, and the HRS harmonization used 2020 cognition for some respondents without a 2022 measure. Third, although items were selected for conceptual comparability, formal measurement invariance across languages and settings was not established. Translation, interviewer procedures, test familiarity, literacy, and cultural differences may affect scores. Lastly, chronic disease, depression, vision, and hearing were based on harmonized self-reported survey reports rather than clinical examinations.

The findings have implications for research and policy. Investments in educational access and quality are likely to have long-term relevance for cognitive health, but policies cannot alter the schooling histories of today's older adults. However, nearer-term strategies can address modifiable conditions such as depression, diabetes, and hearing impairment, while ensuring equitable diagnosis and treatment across settings. Future studies should use longitudinal harmonized data to compare cognitive trajectories, test measurement invariance, schooling quantity and quality, and incorporate richer life-course socioeconomic, migration, language, and health-system measures.

Linking cognitive assessments to objective sensory testing and clinical dementia ascertainment would further clarify whether the residual gaps reflect genuine differences in cognitive health or differences in measurement and context.

In conclusion, later life cognitive performance varies substantially across the five different study settings. Education accounted for a meaningful share of the HRS advantage, particularly relative to the Dominican Republic and Mexico, but it did not explain the gaps with Cuba and Puerto Rico. Cross-setting cognitive disparities therefore appear to reflect both unequal distributions of measured life-course and health characteristics and broader contextual or measurement factors that remain unobserved.

Research In Context

Systematic review: Prior cross-national cognitive-aging research documents that late-life cognition is shaped by education, chronic disease, depression, and sensory function

However, nationally representative evidence from the Hispanic Caribbean remains limited, and few studies quantify how these factors explain cognitive gaps across the U.S., the Caribbean, and Mexico.

Interpretation: Using newly available CADAS data harmonized with HRS and Mexico, we find substantial cross-national differences in TICS-27, delayed recall, and animal naming. Education explains a large share of the U.S. vs Dominican Republic and U.S. vs Mexico gaps, while Cuba and Puerto Rico show large unexplained gaps despite relatively high education.

Future directions: Future work should test measurement invariance, incorporate richer life-

course socioeconomic and health-system exposures, and investigate whether depression, sensory impairment, and chronic disease treatment explain cognitive trajectories over time.

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Table 1. Weighted Summary Statistics for Oaxaca Decomposition Sample

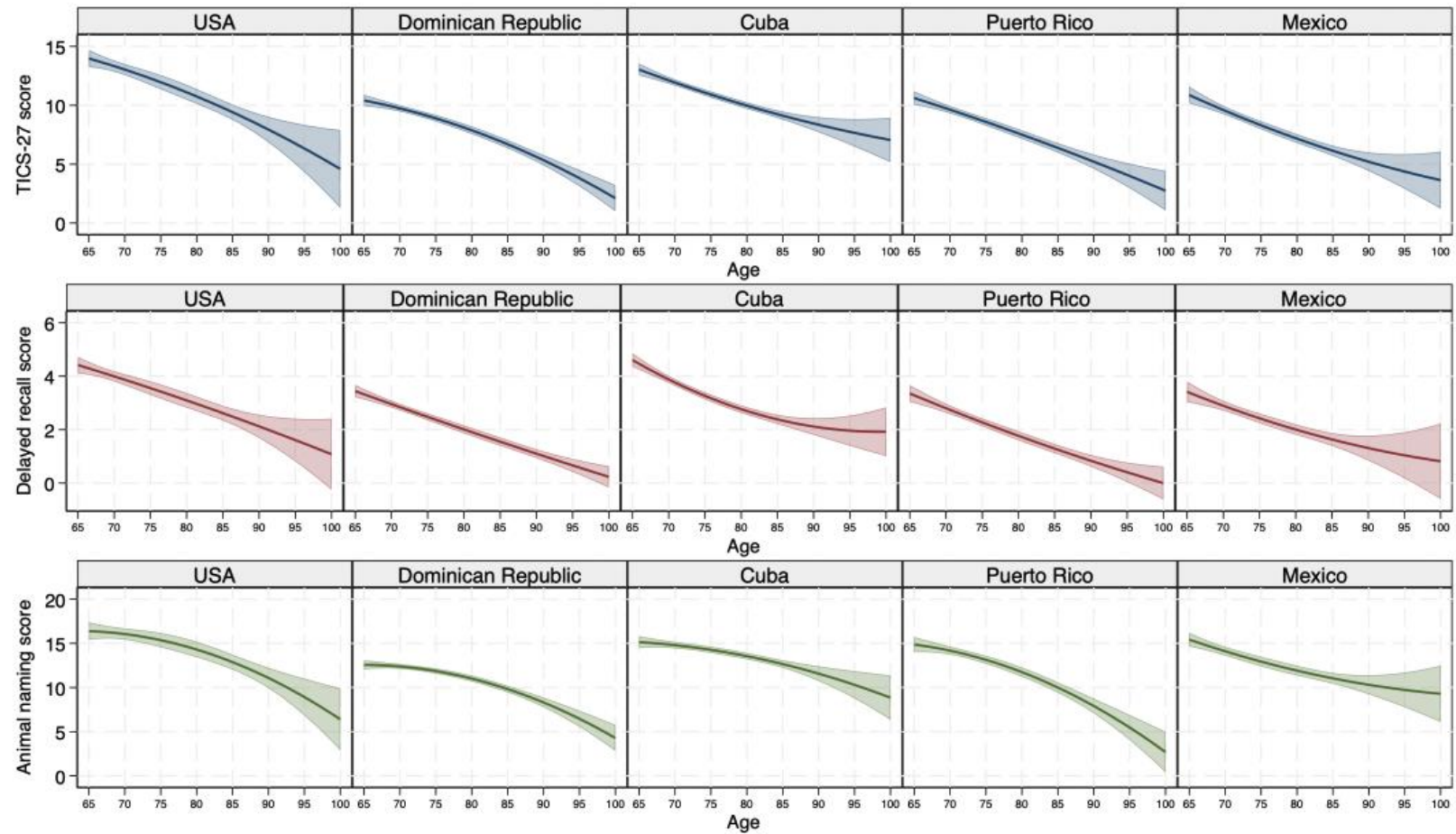
	USA		Dominican Republic		Cuba		Puerto Rico		Mexico	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Demographic characteristics										
Age	73.42	6.93	75.27	7.66	75.88	7.13	75.13	7.07	72.55	6.29
Female	0.58	0.49	0.56	0.50	0.63	0.48	0.65	0.48	0.57	0.49
Education	10.00	4.81	5.09	4.61	11.02	4.21	11.92	4.01	4.87	4.37
Health characteristics										
Diabetes	0.45	0.50	0.26	0.44	0.23	0.47	0.39	0.49	0.29	0.46
Hypertension	0.73	0.44	0.69	0.46	0.70	0.50	0.71	0.45	0.53	0.50
Depression	0.21	0.41	0.29	0.45	0.25	0.43	0.21	0.41	0.45	0.50
Vision										
Excellent	0.06	0.24	0.02	0.13	0.01	0.11	0.37	0.48	0.05	0.21
Very good	0.12	0.32	0.19	0.40	0.06	0.23	0.29	0.45	0.05	0.23
Good	0.43	0.50	0.52	0.50	0.49	0.50	0.20	0.40	0.39	0.49
Fair	0.32	0.47	0.20	0.40	0.35	0.48	0.11	0.31	0.43	0.49
Hearing										
Excellent	0.11	0.31	0.03	0.17	0.03	0.17	0.44	0.50	0.08	0.27
Very good	0.19	0.39	0.27	0.44	0.11	0.31	0.25	0.43	0.07	0.25
Good	0.40	0.49	0.54	0.50	0.59	0.49	0.18	0.39	0.39	0.49
Fair	0.25	0.44	0.13	0.34	0.24	0.42	0.11	0.31	0.41	0.49
Observations	1,063		1,450		1,421		705		2,596	

Notes: Means and standard deviations are weighted using normalized probability weights. The observations row reports unweighted sample sizes in the exact Oaxaca decomposition sample. Means for binary and categorical variables represent weighted proportions. Poor vision and poor hearing are omitted categories.

Table 2. Weighted Cognitive Outcomes by Country

Study source	TICS-27 Mean (SD)	Delayed recall Mean (SD)	Animal naming Mean (SD)	Observations
USA	12.388 (4.844)	3.760 (2.044)	15.500 (6.277)	1,063
Dominican Republic	8.644 (4.630)	2.477 (2.161)	11.520 (4.663)	1,450
Cuba	10.853 (4.156)	3.320 (2.148)	13.997 (5.392)	1,421
Puerto Rico	8.579 (3.864)	2.374 (1.958)	13.150 (5.328)	705
Mexico	9.078 (4.040)	2.740 (2.017)	13.754 (4.979)	2,596

Figure1. Cognitive Outcomes by Age and Country



Lines show weighted quadratic predictions by age. Shaded regions indicate 95% confidence intervals.

Figure 2 : Study Design Oaxaca-Blinder Decomposition

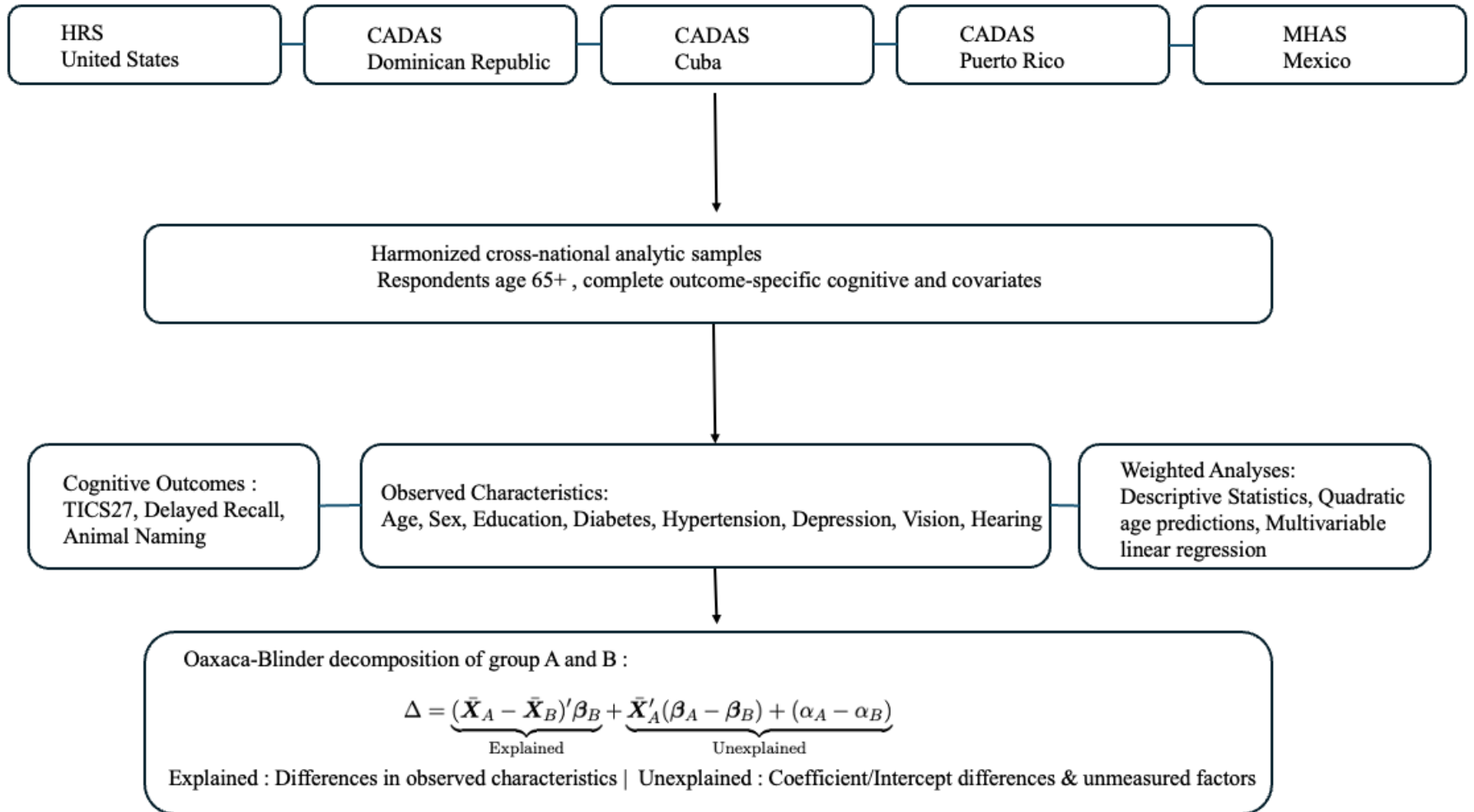


Table 3. Multivariable Linear Regression Results by Cognitive Outcome

Variable	TICS-27		Delayed recall		Animal naming	
	USA	CADAS/MX	USA	CADAS/MX	USA	CADAS/MX
Age	-0.175*** (0.022)	-0.177*** (0.009)	-0.071*** (0.010)	-0.084*** (0.005)	-0.126*** (0.033)	-0.175*** (0.013)
Female	0.202 (0.352)	0.024 (0.135)	0.317** (0.156)	0.580*** (0.074)	-0.215 (0.519)	-0.652*** (0.198)
Education	0.329*** (0.035)	0.301*** (0.014)	0.105*** (0.016)	0.085*** (0.008)	0.362*** (0.050)	0.238*** (0.020)
Diabetes	-0.214 (0.370)	-0.521*** (0.147)	-0.279* (0.163)	-0.374*** (0.083)	-0.711 (0.523)	-0.320 (0.220)
Hypertension	-0.321 (0.420)	0.083 (0.147)	-0.168 (0.162)	0.027 (0.078)	-1.067* (0.639)	-0.313 (0.197)
Depression	-1.406*** (0.425)	-0.755*** (0.145)	-0.532** (0.217)	-0.353*** (0.082)	-0.784 (0.699)	-0.775*** (0.227)
Vision: Excellent	-1.064 (0.871)	-0.055 (0.353)	-0.310 (0.372)	-0.081 (0.217)	0.460 (1.410)	0.900 (0.662)
Vision: Very good	-0.024 (0.785)	0.122 (0.277)	-0.065 (0.316)	-0.123 (0.148)	-0.793 (1.083)	0.930** (0.426)
Vision: Good	0.001 (0.635)	0.657*** (0.227)	-0.171 (0.259)	0.081 (0.116)	-1.292 (0.857)	0.934*** (0.359)
Vision: Fair	-0.093 (0.605)	1.091*** (0.237)	-0.124 (0.257)	0.359*** (0.118)	-1.402 (0.853)	0.581 (0.381)
Hearing: Excellent	1.986** (0.885)	0.228 (0.434)	0.718* (0.423)	-0.002 (0.236)	0.242 (1.275)	0.591 (0.644)
Hearing: Very good	2.226*** (0.786)	1.641*** (0.367)	0.778* (0.398)	0.700*** (0.188)	0.974 (1.194)	0.757 (0.513)
Hearing: Good	2.325*** (0.710)	1.182*** (0.311)	0.903** (0.376)	0.447*** (0.164)	1.078 (1.078)	0.803* (0.470)

Hearing: Fair	1.014 (0.711)	1.722*** (0.321)	0.539 (0.372)	0.685*** (0.172)	0.047 (1.076)	0.823 (0.500)
Constant	20.742*** (1.952)	18.792*** (0.850)	7.535*** (0.878)	7.653*** (0.456)	22.848*** (2.809)	23.940*** (1.164)
Observations	1,063	6,172	1,068	6,245	1,061	6,145
R-squared	0.293	0.266	0.203	0.170	0.172	0.159

Table 4. Oaxaca-Blinder decomposition of cognitive outcomes

Variable	TICS-27		Delayed recall		Animal naming	
	HRS vs CADAS/MX	CADAS vs MX	HRS vs CADAS/MX	CADAS vs MX	HRS vs CADAS/MX	CADAS vs MX
Total observations	7,235	6,172	7,313	6,245	7,206	6,145
Group 1 observations	1,063	3,576	1,068	3,633	1,061	3,533
Group 2 observations	6,172	2,596	6,245	2,612	6,145	2,612
Panel A. Overall decomposition						
Mean: Group 1	12.388 (0.203)	9.509 (0.074)	3.753 (0.086)	2.763 (0.036)	15.406 (0.283)	12.767 (0.089)
Mean: Group 2	9.327 (0.078)	9.078 (0.161)	2.751 (0.041)	2.735 (0.083)	13.189 (0.109)	13.756 (0.222)
Difference	3.061*** (0.218)	0.431** (0.177)	1.002*** (0.095)	0.028 (0.090)	2.217*** (0.303)	-0.989*** (0.239)
Explained	1.100*** (0.130)	0.768*** (0.136)	0.319*** (0.052)	0.128** (0.063)	0.907*** (0.156)	1.115*** (0.160)
Unexplained	1.961*** (0.190)	-0.337** (0.171)	0.683*** (0.092)	-0.100 (0.092)	1.310*** (0.298)	-2.103*** (0.269)
Panel B. Explained component						
Age	0.145** (0.056)	-0.496*** (0.059)	0.067*** (0.025)	-0.235*** (0.029)	0.116** (0.049)	-0.418*** (0.059)
Female	-0.001 (0.003)	0.002 (0.005)	-0.004 (0.011)	0.019 (0.014)	0.004 (0.011)	-0.020 (0.016)
Education	0.904*** (0.094)	1.290*** (0.097)	0.271*** (0.034)	0.360*** (0.044)	0.864*** (0.106)	1.059*** (0.136)

Variable	TICS-27		Delayed recall		Animal naming	
	HRS vs CADAS/MX	CADAS vs MX	HRS vs CADAS/MX	CADAS vs MX	HRS vs CADAS/MX	CADAS vs MX
Diabetes	-0.063* (0.035)	0.010 (0.013)	-0.056*** (0.018)	0.007 (0.009)	-0.090* (0.051)	0.006 (0.010)
Hypertension	-0.012 (0.023)	0.017 (0.027)	-0.007 (0.010)	0.004 (0.015)	-0.074* (0.039)	-0.018 (0.038)
Depression	0.138*** (0.037)	0.143*** (0.034)	0.056*** (0.017)	0.071*** (0.018)	0.102** (0.051)	0.212*** (0.050)
Vision: Excellent	0.004 (0.008)	-0.003 (0.014)	0.001 (0.003)	-0.008 (0.010)	-0.003 (0.010)	0.031 (0.027)
Vision: Very good	0.000 (0.001)	0.004 (0.031)	0.000 (0.001)	-0.018 (0.018)	0.000 (0.002)	0.112** (0.051)
Vision: Good	0.004 (0.009)	0.036* (0.020)	-0.001 (0.002)	0.003 (0.007)	-0.002 (0.008)	0.050* (0.029)
Vision: Fair	-0.002 (0.011)	-0.203*** (0.051)	-0.001 (0.003)	-0.064*** (0.023)	0.003 (0.010)	-0.079 (0.072)
Hearing: Excellent	0.014 (0.021)	0.016 (0.017)	0.004 (0.007)	0.000 (0.008)	0.006 (0.013)	0.029 (0.025)
Hearing: Very good	0.078** (0.039)	0.267*** (0.064)	0.029* (0.016)	0.115*** (0.029)	0.036 (0.031)	0.214*** (0.077)
Hearing: Good	-0.088* (0.047)	0.116*** (0.041)	-0.032* (0.019)	0.043** (0.018)	-0.049 (0.038)	0.125** (0.054)
Hearing: Fair	-0.020 (0.030)	-0.430*** (0.087)	-0.009 (0.014)	-0.170*** (0.042)	-0.006 (0.013)	-0.187* (0.113)

Table 5. Oaxaca-Blinder Decomposition of Cognitive Outcomes by HRS Hispanic Nativity

	HRS Foreign-Born Hispanics vs. HRS U.S.-Born Hispanics			HRS Foreign-Born Hispanics vs. CADAS/Mexico		
	TICS-27	Delayed recall	Animal naming	TICS-27	Delayed recall	Animal naming
Total observations	1,062	1,067	1,060	6,895	6,970	6,870
Group 1 observations	723	725	725	723	725	725
Group 2 observations	339	342	335	6,172	6,245	6,145
Panel A. Overall decomposition						
Mean: Group 1	12.197*** (0.241)	3.709*** (0.101)	14.850*** (0.348)	12.197*** (0.241)	3.709*** (0.101)	14.850*** (0.348)
Mean: Group 2	12.701*** (0.364)	3.825*** (0.154)	16.341*** (0.475)	9.327*** (0.078)	2.751*** (0.041)	13.189*** (0.109)
Difference	-0.504 (0.437)	-0.115 (0.184)	-1.491** (0.589)	2.870*** (0.253)	0.958*** (0.109)	1.661*** (0.364)
Explained	-1.432*** (0.311)	-0.432*** (0.125)	-1.264*** (0.361)	0.793*** (0.150)	0.224*** (0.058)	0.573*** (0.173)
Unexplained	0.928** (0.393)	0.316* (0.184)	-0.227 (0.575)	2.077*** (0.221)	0.734*** (0.106)	1.088*** (0.357)
Panel B. Explained component						
Age	0.110 (0.093)	0.043 (0.036)	0.074 (0.075)	0.199*** (0.069)	0.092*** (0.031)	0.145** (0.057)
Female	0.004 (0.026)	0.002 (0.018)	-0.000 (0.004)	0.000 (0.001)	-0.003 (0.014)	0.005 (0.019)
Education	-1.349*** (0.214)	-0.479*** (0.087)	-1.377*** (0.267)	0.513*** (0.102)	0.142*** (0.031)	0.456*** (0.101)
Diabetes	-0.000 (0.009)	0.001 (0.010)	0.008 (0.041)	-0.063 (0.039)	-0.069*** (0.022)	-0.049 (0.061)

	HRS Foreign-Born Hispanics vs. HRS U.S.-Born Hispanics			HRS Foreign-Born Hispanics vs. CADAS/Mexico		
	TICS-27	Delayed recall	Animal naming	TICS-27	Delayed recall	Animal naming
Hypertension	0.003 (0.011)	0.003 (0.006)	0.032 (0.056)	-0.017 (0.024)	-0.006 (0.010)	-0.055 (0.046)
Depression	-0.044 (0.049)	-0.015 (0.019)	-0.028 (0.039)	0.143*** (0.045)	0.060*** (0.020)	0.099* (0.054)
Vision: Excellent	-0.020 (0.054)	0.000 (0.024)	0.098 (0.107)	-0.011 (0.017)	-0.004 (0.007)	0.016 (0.023)
Vision: Very good	-0.021 (0.053)	0.005 (0.022)	0.078 (0.083)	0.009 (0.014)	0.004 (0.006)	-0.008 (0.017)
Vision: Good	-0.083 (0.127)	0.027 (0.057)	0.314* (0.190)	-0.025 (0.028)	-0.003 (0.011)	-0.016 (0.037)
Vision: Fair	0.004 (0.146)	-0.020 (0.067)	-0.394* (0.203)	0.039 (0.032)	0.011 (0.013)	0.004 (0.040)
Hearing: Excellent	0.061 (0.057)	0.026 (0.026)	0.020 (0.058)	0.031 (0.030)	0.011 (0.013)	0.005 (0.025)
Hearing: Very good	-0.057 (0.068)	-0.020 (0.027)	-0.026 (0.045)	0.049 (0.042)	0.020 (0.018)	0.022 (0.028)
Hearing: Good	-0.136 (0.109)	-0.060 (0.048)	-0.079 (0.088)	-0.096* (0.050)	-0.040 (0.027)	-0.052 (0.056)
Hearing: Fair	0.097 (0.079)	0.056 (0.045)	0.015 (0.103)	0.022 (0.027)	0.011 (0.014)	0.001 (0.016)

Appendix

Table 3 A. Multivariable Linear Regression Results by Cognitive Outcome (Full Table)

Variable	Pooled	USA	Dominican Republic	Cuba	Puerto Rico	Mexico
Panel A. TICS-27						
Age	-0.182*** (0.009)	-0.175*** (0.022)	-0.172*** (0.015)	-0.166*** (0.016)	-0.170*** (0.020)	-0.163*** (0.019)
Female	0.037 (0.130)	0.202 (0.352)	-0.313 (0.219)	-0.338 (0.216)	-0.138 (0.262)	0.375 (0.260)
Education	0.331*** (0.013)	0.329*** (0.035)	0.277*** (0.026)	0.205*** (0.024)	0.270*** (0.035)	0.398*** (0.033)
Diabetes	-0.293** (0.139)	-0.214 (0.370)	-0.079 (0.246)	-0.524*** (0.187)	-0.445* (0.264)	-0.458 (0.298)
Hypertension	0.044 (0.140)	-0.321 (0.420)	-0.070 (0.235)	0.370* (0.222)	0.303 (0.294)	0.044 (0.271)
Depression	-0.917*** (0.139)	-1.406*** (0.425)	-1.207*** (0.232)	-0.173 (0.230)	-0.638** (0.317)	-0.801*** (0.261)
Vision: Excellent	-0.431 (0.350)	-1.064 (0.871)	1.430 (0.916)	1.606 (0.990)	0.928 (0.622)	0.107 (0.658)
Vision: Very good	-0.008 (0.279)	-0.024 (0.785)	0.793 (0.501)	1.837*** (0.566)	1.347** (0.616)	-0.541 (0.518)
Vision: Good	0.584** (0.228)	0.001 (0.635)	0.636 (0.439)	0.963*** (0.353)	1.303** (0.657)	0.530 (0.401)
Vision: Fair	0.928*** (0.232)	-0.093 (0.605)	1.995*** (0.493)	1.554*** (0.363)	0.534 (0.664)	0.530 (0.378)
Hearing: Excellent	0.438 (0.414)	1.986** (0.885)	3.218*** (0.908)	0.850 (0.878)	1.060 (0.854)	-0.105 (0.721)
Hearing: Very good	1.704*** (0.335)	2.226*** (0.786)	3.341*** (0.760)	1.166 (0.718)	0.819 (0.858)	2.246*** (0.731)
Hearing: Good	1.207*** (0.288)	2.325*** (0.710)	2.714*** (0.733)	0.398 (0.657)	1.288 (0.868)	0.931** (0.437)

Variable	Pooled	USA	Dominican Republic	Cuba	Puerto Rico	Mexico
Hearing: Fair	1.536*** (0.294)	1.014 (0.711)	3.282*** (0.777)	1.691** (0.673)	0.932 (0.892)	1.265*** (0.427)
Observations	7,235	1,063	1,450	1,421	705	2,596
R-squared	0.281	0.293	0.267	0.186	0.264	0.344
Panel B. Delayed recall						
Age	-0.084*** (0.005)	-0.071*** (0.010)	-0.086*** (0.007)	-0.096*** (0.008)	-0.092*** (0.010)	-0.067*** (0.011)
Female	0.536*** (0.068)	0.317** (0.156)	0.395*** (0.105)	0.237** (0.114)	0.517*** (0.133)	0.836*** (0.151)
Education	0.096*** (0.007)	0.105*** (0.016)	0.083*** (0.012)	0.036*** (0.012)	0.075*** (0.017)	0.109*** (0.020)
Diabetes	-0.298*** (0.075)	-0.279* (0.163)	-0.199* (0.115)	-0.351*** (0.100)	-0.184 (0.138)	-0.389** (0.179)
Hypertension	0.017 (0.072)	-0.168 (0.162)	-0.104 (0.115)	0.111 (0.098)	0.063 (0.151)	0.030 (0.157)
Depression	-0.393*** (0.077)	-0.532** (0.217)	-0.615*** (0.108)	-0.121 (0.123)	-0.070 (0.177)	-0.395*** (0.149)
Vision: Excellent	-0.207 (0.194)	-0.310 (0.372)	0.627 (0.511)	0.843 (0.590)	0.805*** (0.271)	-0.415 (0.456)
Vision: Very good	-0.144 (0.138)	-0.065 (0.316)	0.166 (0.245)	0.568* (0.304)	0.717*** (0.273)	-0.457 (0.311)
Vision: Good	0.055 (0.109)	-0.171 (0.259)	0.130 (0.207)	0.241 (0.193)	0.628** (0.282)	-0.060 (0.206)
Vision: Fair	0.292*** (0.110)	-0.124 (0.257)	0.793*** (0.222)	0.470** (0.199)	0.210 (0.294)	0.100 (0.190)
Hearing: Excellent	0.088 (0.213)	0.718* (0.423)	0.812* (0.420)	0.523 (0.498)	0.442 (0.513)	-0.113 (0.377)
Hearing: Very good	0.701*** (0.172)	0.778* (0.398)	0.751** (0.310)	0.492 (0.382)	0.607 (0.513)	1.235*** (0.331)
Hearing: Good	0.468*** (0.152)	0.903** (0.376)	0.451 (0.290)	0.180 (0.345)	0.613 (0.514)	0.525** (0.241)

Variable	Pooled	USA	Dominican Republic	Cuba	Puerto Rico	Mexico
Hearing: Fair	0.633*** (0.159)	0.539 (0.372)	0.757** (0.307)	0.853** (0.352)	0.716 (0.520)	0.566** (0.239)
Observations	7,313	1,068	1,465	1,441	727	2,612
R-squared	0.179	0.203	0.203	0.148	0.206	0.190
Panel C. Animal naming						
Age	-0.172*** (0.012)	-0.126*** (0.033)	-0.112*** (0.016)	-0.102*** (0.020)	-0.209*** (0.028)	-0.161*** (0.026)
Female	-0.573*** (0.189)	-0.215 (0.519)	-0.894*** (0.231)	-0.814*** (0.294)	-0.253 (0.392)	-0.603 (0.429)
Education	0.271*** (0.019)	0.362*** (0.050)	0.211*** (0.027)	0.369*** (0.032)	0.267*** (0.049)	0.209*** (0.062)
Diabetes	-0.282 (0.205)	-0.711 (0.523)	-0.043 (0.253)	0.032 (0.276)	-1.170*** (0.401)	-0.260 (0.492)
Hypertension	-0.414** (0.189)	-1.067* (0.639)	0.300 (0.250)	-0.152 (0.219)	0.705 (0.438)	-0.349 (0.402)
Depression	-0.817*** (0.219)	-0.784 (0.699)	-0.863*** (0.242)	-0.545* (0.299)	-0.750 (0.481)	-1.463*** (0.407)
Vision: Excellent	0.765 (0.606)	0.460 (1.410)	0.778 (0.922)	0.845 (1.315)	1.264 (1.232)	1.441 (1.326)
Vision: Very good	0.662 (0.406)	-0.793 (1.083)	1.360** (0.652)	1.721** (0.763)	1.349 (1.215)	1.251 (0.886)
Vision: Good	0.647* (0.338)	-1.292 (0.857)	0.907 (0.555)	0.943* (0.543)	0.588 (1.239)	1.016* (0.616)
Vision: Fair	0.311 (0.356)	-1.402 (0.853)	1.478** (0.581)	0.569 (0.547)	0.355 (1.292)	-0.060 (0.670)
Hearing: Excellent	0.383 (0.581)	0.242 (1.275)	7.920*** (0.948)	1.812 (1.104)	-0.372 (1.665)	-0.814 (1.078)
Hearing: Very good	0.795* (0.475)	0.974 (1.194)	6.225*** (0.815)	1.063 (0.961)	-0.715 (1.670)	0.815 (0.944)
Hearing: Good	0.720* (0.431)	1.078 (1.078)	5.177*** (0.749)	0.944 (0.859)	-0.490 (1.689)	0.163 (0.711)

Variable	Pooled	USA	Dominican Republic	Cuba	Puerto Rico	Mexico
Hearing: Fair	0.618 (0.457)	0.047 (1.076)	4.183*** (0.786)	-0.346 (0.874)	-1.020 (1.727)	0.297 (0.716)
Observations	7,206	1,061	1,444	1,418	671	2,612
R-squared	0.166	0.172	0.254	0.157	0.203	0.172

Table 4 A : Oaxaca-Blinder Decomposition of Cognitive outcomes : Unexplained component

Variable	TICS-27		Delayed recall		Animal naming	
	HRS vs DR/CU/PR/MX	DR/CU/PR vs MX	HRS vs DR/CU/PR/MX	DR/CU/PR vs MX	HRS vs DR/CU/PR/MX	DR/CU/PR vs MX
Panel C. Unexplained component						
Age	0.176 (1.757)	-0.902 (1.592)	0.957 (0.824)	-1.899** (0.883)	3.645 (2.604)	2.652 (2.133)
Female	0.104 (0.220)	-0.380** (0.173)	-0.155 (0.101)	-0.271*** (0.098)	0.257 (0.325)	-0.033 (0.271)
Education	0.237 (0.325)	-0.942*** (0.244)	0.174 (0.150)	-0.234* (0.140)	1.063** (0.454)	0.863** (0.435)
Diabetes	0.113 (0.145)	-0.030 (0.093)	0.035 (0.067)	0.009 (0.054)	-0.144 (0.208)	-0.044 (0.149)
Hypertension	-0.275 (0.301)	0.068 (0.191)	-0.133 (0.122)	-0.011 (0.106)	-0.514 (0.454)	0.301 (0.266)
Depression	-0.178 (0.122)	0.037 (0.106)	-0.049 (0.063)	0.021 (0.059)	-0.002 (0.199)	0.255 (0.157)
Vision: Excellent	-0.066 (0.061)	-0.024 (0.049)	-0.015 (0.028)	0.028 (0.032)	-0.029 (0.100)	-0.074 (0.091)
Vision: Very good	-0.017 (0.095)	0.122* (0.063)	0.007 (0.040)	0.060 (0.037)	-0.196 (0.132)	-0.030 (0.102)
Vision: Good	-0.280 (0.286)	0.089 (0.199)	-0.108 (0.121)	0.101 (0.102)	-0.954** (0.397)	-0.098 (0.300)
Vision: Fair	-0.379* (0.207)	0.382** (0.159)	-0.155* (0.090)	0.165** (0.079)	-0.635** (0.298)	0.332 (0.260)

Variable	TICS-27		Delayed recall		Animal naming	
	HRS vs DR/CU/PR/MX	DR/CU/PR vs MX	HRS vs DR/CU/PR/MX	DR/CU/PR vs MX	HRS vs DR/CU/PR/MX	DR/CU/PR vs MX
Panel C. Unexplained component						
Hearing: Excellent	0.182* (0.102)	0.105 (0.083)	0.074 (0.050)	0.023 (0.042)	-0.036 (0.145)	0.312*** (0.120)
Hearing: Very good	0.097 (0.143)	-0.064 (0.117)	0.013 (0.072)	-0.096* (0.054)	0.036 (0.212)	0.219 (0.152)
Hearing: Good	0.484 (0.326)	0.247 (0.276)	0.194 (0.173)	-0.065 (0.141)	0.116 (0.495)	0.983** (0.412)
Hearing: Fair	-0.185 (0.203)	0.312* (0.182)	-0.038 (0.107)	0.089 (0.093)	-0.204 (0.310)	0.284 (0.271)
Constant	1.949 (2.116)	0.644 (1.893)	-0.118 (0.984)	1.980* (1.008)	-1.092 (3.022)	-8.026*** (2.454)

Table 4 B. Oaxaca-Blinder Decomposition of TICS-27 Scores

	HRS vs DR	HRS vs CU	HRS vs PR	HRS vs Mexico
Total observations	2,513	2,484	1,768	3,659
Group 1 observations	1,063	1,063	1,063	1,063
Group 2 observations	1,450	1,421	705	2,596
Mean: Group 1	12.388 (0.203)	12.388 (0.203)	12.388 (0.203)	12.388 (0.203)
Mean: Group 2	8.644 (0.122)	10.853 (0.110)	8.579 (0.145)	9.078 (0.161)
Difference	3.744*** (0.237)	1.535*** (0.231)	3.809*** (0.250)	3.310*** (0.259)
Explained	1.811*** (0.177)	0.122 (0.133)	-0.262 (0.210)	1.993*** (0.206)
Unexplained	1.933*** (0.240)	1.413*** (0.210)	4.071*** (0.273)	1.317*** (0.252)
Age	0.320*** (0.064)	0.419*** (0.066)	0.295*** (0.071)	-0.147** (0.066)
Female	-0.001 (0.004)	0.003 (0.010)	-0.002 (0.014)	0.003 (0.009)
Education	1.489*** (0.131)	-0.271*** (0.069)	-0.573*** (0.093)	1.866*** (0.162)
Diabetes	-0.028 (0.043)	-0.082* (0.047)	-0.021 (0.017)	-0.054 (0.039)
Hypertension	-0.007 (0.010)	0.001 (0.007)	-0.000 (0.004)	-0.028 (0.051)
Depression	0.101*** (0.034)	0.033 (0.020)	-0.002 (0.025)	0.263*** (0.068)

Vision: Excellent	0.008 (0.028)	0.014 (0.033)	0.021 (0.164)	-0.007 (0.011)
Vision: Very good	-0.030 (0.036)	0.054* (0.031)	-0.113 (0.086)	-0.017 (0.029)
Vision: Good	-0.027 (0.033)	-0.030 (0.026)	0.153 (0.108)	0.012 (0.019)
Vision: Fair	0.109** (0.049)	-0.026 (0.021)	0.046 (0.093)	-0.024 (0.040)
Hearing: Excellent	0.203*** (0.066)	0.112** (0.054)	-0.500** (0.204)	0.031 (0.028)
Hearing: Very good	-0.230*** (0.073)	0.129** (0.052)	-0.097* (0.052)	0.262*** (0.078)
Hearing: Good	-0.359*** (0.097)	-0.258*** (0.098)	0.389*** (0.129)	0.008 (0.050)
Hearing: Fair	0.263*** (0.078)	0.024 (0.031)	0.144* (0.086)	-0.173** (0.071)
Age	-0.250 (1.967)	-0.654 (2.018)	-0.360 (2.188)	-0.863 (2.128)
Female	0.294 (0.236)	0.327 (0.249)	0.209 (0.267)	-0.100 (0.251)
Education	0.391 (0.328)	1.302*** (0.445)	0.645 (0.543)	-0.513 (0.361)
Diabetes	-0.048 (0.157)	0.106 (0.141)	0.097 (0.190)	0.091 (0.176)
Hypertension	-0.179 (0.341)	-0.496 (0.339)	-0.452 (0.368)	-0.230 (0.314)
Depression	-0.049 (0.119)	-0.283** (0.111)	-0.160 (0.110)	-0.199 (0.163)
Vision: Excellent	-0.099* (0.052)	-0.098* (0.051)	-0.430* (0.230)	-0.064 (0.060)
Vision: Very good	-0.126 (0.143)	-0.160* (0.084)	-0.276 (0.200)	0.044 (0.080)

Vision: Good	-0.302 (0.364)	-0.447 (0.335)	-0.411 (0.287)	-0.218 (0.307)
Vision: Fair	-0.545*** (0.204)	-0.553** (0.236)	-0.134 (0.190)	-0.233 (0.265)
Hearing: Excellent	-0.087 (0.090)	0.080 (0.088)	0.254 (0.334)	0.195* (0.108)
Hearing: Very good	-0.253 (0.247)	0.157 (0.157)	0.306 (0.252)	-0.003 (0.136)
Hearing: Good	-0.183 (0.476)	0.950** (0.475)	0.301 (0.323)	0.551* (0.328)
Hearing: Fair	-0.438** (0.204)	-0.166 (0.239)	0.015 (0.204)	-0.083 (0.273)
Constant	3.806 (2.515)	1.348 (2.490)	4.467 (2.759)	2.940 (2.522)

Table 4 C. Oaxaca-Blinder Decomposition of Delayed Recall Scores

	HRS vs DR	HRS vs CU	HRS vs PR	HRS vs Mexico
Total observations	2,533	2,509	1,795	3,680
Group 1 observations	1,068	1,068	1,068	1,068
Group 2 observations	1,465	1,441	727	2,612
Mean delayed recall: group 1	3.753 (0.086)	3.753 (0.086)	3.753 (0.086)	3.753 (0.086)
Mean delayed recall: group 2	2.452 (0.057)	3.307 (0.057)	2.311 (0.073)	2.735 (0.083)
Difference	1.301*** (0.103)	0.446*** (0.103)	1.442*** (0.113)	1.018*** (0.119)
Explained	0.637*** (0.076)	0.091 (0.057)	-0.118 (0.088)	0.561*** (0.089)
Unexplained	0.664*** (0.115)	0.354*** (0.102)	1.560*** (0.123)	0.457*** (0.129)
Age	0.151*** (0.029)	0.206*** (0.032)	0.143*** (0.034)	-0.059** (0.027)
Female	0.008 (0.009)	-0.013 (0.008)	-0.026* (0.013)	0.006 (0.018)
Education	0.463*** (0.054)	-0.071*** (0.020)	-0.170*** (0.032)	0.549*** (0.071)
Diabetes	-0.046** (0.020)	-0.070*** (0.023)	-0.015 (0.009)	-0.054** (0.022)
Hypertension	-0.006 (0.005)	-0.001 (0.003)	-0.001 (0.003)	-0.014 (0.023)
Depression	0.045*** (0.016)	0.012 (0.008)	-0.001 (0.007)	0.111*** (0.034)
Vision: Excellent	0.007 (0.014)	0.013 (0.018)	-0.075 (0.069)	-0.006 (0.007)
Vision: Very good	-0.004 (0.015)	0.015 (0.013)	-0.056 (0.036)	-0.016 (0.014)
Vision: Good	0.002 (0.013)	-0.002 (0.010)	0.053 (0.045)	-0.005 (0.008)

	HRS vs DR	HRS vs CU	HRS vs PR	HRS vs Mexico
Vision: Fair	0.037* (0.020)	-0.006 (0.007)	0.009 (0.040)	0.001 (0.018)
Hearing: Excellent	0.059** (0.026)	0.049* (0.028)	-0.191* (0.109)	0.010 (0.011)
Hearing: Very good	-0.062** (0.026)	0.048** (0.024)	-0.044* (0.026)	0.117*** (0.036)
Hearing: Good	-0.093** (0.037)	-0.103** (0.050)	0.163** (0.071)	0.005 (0.022)
Hearing: Fair	0.077** (0.031)	0.014 (0.016)	0.093* (0.049)	-0.084** (0.037)
Age	1.118 (0.888)	1.827* (0.973)	1.504 (1.035)	-0.342 (1.078)
Female	-0.044 (0.107)	0.049 (0.116)	-0.123 (0.125)	-0.300** (0.125)
Education	0.165 (0.150)	0.719*** (0.210)	0.326 (0.252)	-0.028 (0.186)
Diabetes	-0.029 (0.071)	0.025 (0.065)	-0.040 (0.089)	0.041 (0.090)
Hypertension	-0.046 (0.140)	-0.201 (0.135)	-0.167 (0.159)	-0.125 (0.141)
Depression	0.021 (0.060)	-0.094* (0.057)	-0.096* (0.058)	-0.045 (0.086)
Vision: Excellent	-0.037 (0.025)	-0.042 (0.026)	-0.237** (0.098)	0.006 (0.032)
Vision: Very good	-0.035 (0.061)	-0.054 (0.038)	-0.157* (0.084)	0.033 (0.037)
Vision: Good	-0.143 (0.156)	-0.192 (0.150)	-0.255** (0.121)	-0.046 (0.136)
Vision: Fair	-0.241*** (0.089)	-0.199* (0.108)	-0.072 (0.083)	-0.084 (0.119)
Hearing: Excellent	-0.007 (0.042)	0.014 (0.045)	0.075 (0.180)	0.077 (0.053)

	HRS vs DR	HRS vs CU	HRS vs PR	HRS vs Mexico
Hearing: Very good	0.006 (0.113)	0.042 (0.081)	0.037 (0.139)	-0.058 (0.065)
Hearing: Good	0.212 (0.221)	0.359 (0.251)	0.085 (0.185)	0.150 (0.176)
Hearing: Fair	-0.043 (0.094)	-0.077 (0.125)	-0.032 (0.115)	-0.009 (0.145)
Constant	-0.234 (1.115)	-1.819 (1.199)	0.711 (1.319)	1.188 (1.245)

Table 4 D. Oaxaca - Blinder Decomposition of Animal Naming Scores

	HRS vs DR	HRS vs CU	HRS vs PR	HRS vs Mexico
Total observations	2,505	2,479	1,732	3,673
Group 1 observations	1,061	1,061	1,061	1,061
Group 2 observations	1,444	1,418	671	2,612
Mean: Group 1	15.406*** (0.283)	15.406*** (0.283)	15.406*** (0.283)	15.406*** (0.283)
Mean: Group 2	11.427*** (0.125)	13.981*** (0.143)	13.085*** (0.208)	13.756*** (0.222)
Difference	3.979*** (0.309)	1.425*** (0.317)	2.321*** (0.351)	1.650*** (0.359)
Explained	1.450*** (0.221)	-0.095 (0.180)	-0.882*** (0.291)	1.548*** (0.260)
Unexplained	2.529*** (0.337)	1.520*** (0.322)	3.204*** (0.416)	0.103 (0.409)
Age	0.221*** (0.053)	0.272*** (0.060)	0.266*** (0.074)	-0.130** (0.058)
Female	-0.012 (0.016)	0.025 (0.019)	0.015 (0.022)	-0.004 (0.013)
Education	1.395*** (0.155)	-0.370*** (0.095)	-0.600*** (0.108)	1.459*** (0.218)
Diabetes	-0.074 (0.057)	-0.077 (0.067)	-0.058* (0.034)	-0.079 (0.060)
Hypertension	-0.016 (0.017)	-0.017 (0.018)	-0.004 (0.009)	-0.147* (0.081)
Depression	0.065* (0.034)	0.027 (0.021)	0.002 (0.019)	0.272*** (0.103)
Vision: Excellent	0.028 (0.039)	0.033 (0.050)	-0.269 (0.290)	0.015 (0.022)

Vision: Very good	-0.021 (0.047)	0.028 (0.040)	-0.048 (0.139)	0.014 (0.043)
Vision: Good	0.015 (0.041)	0.010 (0.031)	-0.085 (0.181)	-0.006 (0.024)
Vision: Fair	0.004 (0.057)	0.015 (0.020)	-0.108 (0.158)	0.083 (0.065)
Hearing: Excellent	0.318*** (0.093)	0.081 (0.068)	0.021 (0.343)	-0.010 (0.029)
Hearing: Very good	-0.289*** (0.094)	0.077 (0.061)	-0.008 (0.063)	0.104 (0.090)
Hearing: Good	-0.441*** (0.122)	-0.195 (0.134)	0.063 (0.214)	0.003 (0.019)
Hearing: Fair	0.257*** (0.092)	-0.003 (0.016)	-0.071 (0.148)	-0.026 (0.097)
Age	-1.019 (2.705)	-1.764 (2.861)	6.171* (3.186)	2.560 (3.065)
Female	0.388 (0.323)	0.364 (0.360)	0.024 (0.397)	0.224 (0.388)
Education	1.140*** (0.426)	-0.066 (0.615)	1.037 (0.756)	1.138* (0.588)
Diabetes	-0.240 (0.207)	-0.255 (0.202)	0.195 (0.278)	-0.169 (0.268)
Hypertension	-0.978** (0.488)	-0.660 (0.485)	-1.285** (0.558)	-0.454 (0.475)
Depression	0.019 (0.181)	-0.054 (0.172)	-0.007 (0.176)	0.223 (0.264)
Vision: Excellent	-0.013 (0.067)	-0.014 (0.071)	-0.176 (0.405)	-0.053 (0.105)
Vision: Very good	-0.329* (0.193)	-0.216* (0.115)	-0.432 (0.326)	-0.174 (0.119)
Vision: Good	-1.044** (0.483)	-1.039** (0.470)	-0.590 (0.469)	-0.952** (0.434)

Vision: Fair	-0.750*** (0.269)	-0.659* (0.337)	-0.375 (0.328)	-0.500 (0.403)
Hearing: Excellent	-0.544*** (0.129)	-0.111 (0.119)	0.169 (0.571)	0.098 (0.155)
Hearing: Very good	-1.182*** (0.328)	-0.013 (0.224)	0.365 (0.440)	0.020 (0.192)
Hearing: Good	-1.919*** (0.613)	0.066 (0.677)	0.454 (0.575)	0.362 (0.508)
Hearing: Fair	-0.807*** (0.262)	0.096 (0.337)	0.194 (0.367)	-0.083 (0.425)
Constant	9.806*** (3.234)	5.847* (3.441)	-2.541 (4.259)	-2.137 (3.509)

Table 5 A : Unexplained Components of the Oaxaca-Blinder Decomposition (Nativity analysis)

	HRS Foreign-Born Hispanics vs. HRS U.S.-Born Hispanics			HRS Foreign-Born Hispanics vs. CADAS/Mexico		
	TICS-27	Delayed recall	Animal naming	TICS-27	Delayed recall	Animal naming
Panel A. Total unexplained component						
Total unexplained	0.928** (0.393)	0.316* (0.184)	-0.227 (0.575)	2.077*** (0.221)	0.734*** (0.106)	1.088*** (0.357)
Panel B. Detailed unexplained component						
Age	-4.682 (3.504)	-2.555 (1.584)	1.152 (5.423)	-0.679 (2.052)	0.454 (0.959)	4.537 (3.053)
Female	-0.678 (0.445)	-0.051 (0.195)	-0.668 (0.639)	-0.033 (0.244)	-0.139 (0.112)	0.009 (0.390)
Education	-2.555*** (0.858)	-1.438*** (0.415)	-2.743** (1.321)	0.022 (0.328)	-0.006 (0.147)	0.496 (0.501)
Diabetes	-0.013 (0.350)	-0.193 (0.155)	0.556 (0.483)	0.114 (0.160)	-0.019 (0.077)	0.025 (0.259)
Hypertension	-0.432 (0.638)	-0.046 (0.238)	0.729 (0.954)	-0.344 (0.326)	-0.119 (0.145)	-0.332 (0.598)
Depression	-0.200 (0.200)	-0.088 (0.104)	-0.067 (0.306)	-0.277** (0.141)	-0.094 (0.069)	-0.039 (0.232)
Vision: Excellent	-0.128 (0.108)	-0.056 (0.048)	-0.076 (0.198)	-0.111 (0.077)	-0.032 (0.035)	0.011 (0.126)
Vision: Very good	-0.287 (0.193)	-0.035 (0.081)	0.227 (0.281)	-0.099 (0.108)	-0.010 (0.044)	-0.118 (0.147)
Vision: Good	-0.276 (0.554)	0.127 (0.246)	0.974 (0.783)	-0.217 (0.302)	-0.026 (0.126)	-0.533 (0.427)

	HRS Foreign-Born Hispanics vs. HRS U.S.-Born Hispanics			HRS Foreign-Born Hispanics vs. CADAS/Mexico		
	TICS-27	Delayed recall	Animal naming	TICS-27	Delayed recall	Animal naming
Vision: Fair	-0.079 (0.378)	-0.002 (0.173)	0.749 (0.502)	-0.438* (0.262)	-0.165 (0.114)	-0.381 (0.385)
Hearing: Excellent	0.101 (0.187)	0.037 (0.089)	-0.155 (0.254)	0.176 (0.125)	0.081 (0.073)	-0.100 (0.173)
Hearing: Very good	-0.008 (0.291)	-0.023 (0.154)	-0.118 (0.412)	0.024 (0.167)	-0.008 (0.100)	-0.020 (0.232)
Hearing: Good	-0.630 (0.537)	-0.193 (0.307)	-0.477 (0.796)	0.094 (0.370)	0.080 (0.245)	-0.128 (0.536)
Hearing: Fair	-0.389 (0.341)	-0.174 (0.191)	-0.427 (0.469)	-0.435* (0.245)	-0.134 (0.163)	-0.442 (0.354)
Constant	11.182*** (4.263)	5.005** (1.959)	0.117 (6.249)	4.281* (2.469)	0.873 (1.166)	-1.897 (3.450)

